

1. Questions

Read the following information carefully and answer the questions based on it.

The table chart given below shows the total number of employees (marketing + HR) working in four different companies P, Q, R and S. It also shows the ratio of the number of male employees working in marketing and HR departments and the ratio of the number of female employees working marketing and HR department in each company.

Company	The total number of employees working (marketing + HR)	The ratio of the number of male employees working in marketing and HR department	The ratio of the number of female employees working in marketing and HR department
P	450	15:14	5:3
Q	600	26:17	12:5
R	280	32:9	8:7
S	360	15:8	9:4

Note:

i). The total number of employees working = the number of employees working in marketing department + the number of employees working in HR department

ii). The total number of employees working in marketing department = the number of male employees working marketing department + the number of female employees working marketing department

iii). The total number of employees working in HR department = the number of male employees working HR department + the number of female employees working HR department

iv). In company P, 44.44% of the total number of employees is working in HR department.

v). The difference between the total number of employees working in marketing and HR department in company Q is 160. (*marketing* > *HR*)

vi). The total number of employees working in marketing department in company S is 20% more than that of company R.

vii). The total number of employees working in HR department in company R is $\frac{2}{3}$ rd of the total number of employees working in HR department in company S.

In the HR department (under graduate + post graduate) of company S, if ____% of male employees are undergraduates, which is ____ more than the number of female undergraduates, then the ratio of male to female postgraduates will be 2:1.

Which of the following does satisfy the fillers?

D). 75 and 30

II). 55 and 22

III). 60 and 28

- a. Only I follows
- b. Both I and II follow
- c. Both II and III follow
- d. All I, II and III follow
- e. None follows

2. Questions

In the marketing department of company R, the ratio of males who didn't get promotions to females who got promotions is 12:5. If males who got promotions is 85 more than females who didn't get promotions, find the total number of employees who got promotions (male + female).

- a. 150
- b. 110
- c. 125
- d. 145
- e. None of these

3. Questions

In company P, sales department has 50% more employees than HR department. The number of females in accounts department is 60, which is half the number in sales department. If male employees in sales and accounts departments are equal, find the total number of employees (males + females) in accounts department.

- a. 240
- b. 360
- c. 150
- d. Cannot be determined
- e. None of these

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4. Questions

Company T has 66.67% more marketing employees than company S. If $\frac{3}{8}$ of them in marketing department of company T are females. In company T, male HR employees are equal to half of marketing male employees, and the female ratio of marketing to HR is 2:1. Find the total number of employees in marketing and HR in company T.

- a. 480

- b. 560
- c. 720
- d. 600
- e. None of these

5. Questions

In company Q's HR department, $65X$ new employees joined with the ratio of male to female employees, which is 3:10. The number of male employees is $25X$ more than females. The value of $[X]$ is a ____[?]____ number.

What should come in place of question mark?

- a. Odd
- b. Prime
- c. Composite
- d. All of the above
- e. None of these

6. Questions

The following question has three statements as statement I, statement II and statement III. You have to determine whether data provided in three statements are sufficient to answer the question or not.

Find the difference between the initial mixture in vessels A and B.

Statement I: The ratio of the quantity of milk and water in vessels A and B is 2:1 and 3:2 respectively.

Statement II: If vessels A and B are mixed together, then the quantity of milk will be 116 ml more than water in the resultant mixture.

Statement III: The total initial mixture of vessels A and B together is 420 ml.

- a. Both statements I and II together are necessary
- b. Only statement II alone is sufficient
- c. Both statements II and III together are necessary
- d. All statements I, II and III together are necessary
- e. Neither statement I or II nor III is sufficient

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7. Questions

If the marked-price of articles A and B are equal, then find the cost price of article B.

Statement I: The cost price of article A is Rs. [8P], which is sold at a profit of 37.50% after allowing a discount of Rs. 100.

Statement II: The selling price of article B is Rs. 960, which is sold at profit of Rs. $[3P+60]$.

Statement III: Article B has been marked-up by 100% above its cost price.

- a. Both statements I and II together are necessary
- b. Only statement II alone is sufficient
- c. Both statements II and III together are necessary
- d. All statements I, II and III together are necessary
- e. Neither statement I or II nor III is sufficient

8. Questions

If the time taken by train X to cross the platform of length 80 meters is 1 second less than the time taken by train Y to cross the same platform, then find the length of train Y.

Statement I: The length of the train X is 40 meters less than train Y.

Statement II: The time taken by train X to cross a man running with the speed of 20 m/sec in the same direction is 4 seconds. (Speed of train X is more than the speed of the man)

Statement III: The speed of train Y is 80 m/sec, which is 11.11% less than train X.

- a. Either both I and III together or both II and III are necessary
- b. Only statement II alone is sufficient
- c. Both statements I and II together are necessary
- d. All statements I, II and III together are necessary
- e. Neither statement I or II nor III is sufficient

9. Questions

If the amount invested in scheme A (S.I.) is Rs. 32000, which is $\frac{2}{3}$ rd of the amount invested in scheme B (C.I.), then find the interest received from in scheme A after 3 years. (The rate of interest offered in scheme B is a positive integer)

Statement I: The interest received from scheme B after 2 years is Rs. 10080.

Statement II: The ratio of the rate of interest offered in scheme A and B is 5:4.

Statement III: The difference between the rate of interest offered in scheme A and B is 2.50. ($A > B$)

- a. Both statements I and II together are necessary
- b. Only statement II alone is sufficient
- c. Any of the two given statements are sufficient
- d. All statements I, II and III together are necessary
- e. Neither statement I or II nor III is sufficient

10. Questions

Find the time taken by R to complete the whole work alone, if the total work is 60 units.

Statement I: P, Q and R together can complete the whole work in $(60/13)$ days.

Statement II: The ratio of the efficiency of P and R is 3:4.

Statement III: The time taken by Q and R together to complete half of the total work is 3 days.

- a. Both statements I and II together are necessary
- b. Only statement II alone is sufficient
- c. Both statements II and III together are necessary
- d. All statements I, II and III together are necessary
- e. Neither statement I or II nor III is sufficient

11. Questions

Read the following information carefully and answer the questions based on it.

Three flights namely IndiGo, SpiceJet and Vistara, in which some of the seats are booked, out of which some seats are economy class, while the rest of the seats are business class. Out of the total number of economy and business class seats booked, some seats are occupied, while the rest of the seats remained vacant.

IndiGo: The total number of seats booked on the flight is 800. The ratio of the number of economy and business class seats booked is 5:3. The number of economy class seats which are occupied is 4 times of the number of business class seats which remained vacant. The number of business class seats which are occupied is 12.50% more than the number of economy class seats which remained vacant.

SpiceJet: The total number of business class seats booked on the flight is 150, which is $1/3^{\text{rd}}$ of the total number of economy class seats booked in flight. The ratio of the number of economy class seats which are occupied and the number of business class seats which remained vacant is 5:1. The number of business class seats which are occupied is half of the number of economy class seats which remained vacant.

Vistara: The ratio of the number of economy class seats which are occupied and vacant is 2:1. The number of business class seats which are vacant is 14.28% of the number of business class seats which are occupied. The number of seats which are occupied in economy class is 50 more than that of business class. The number of seats which remained vacant in business class is 150 less than that of economy class.

Note:

- i). The total number of seats booked = the number of economy class seats booked + the number of business class seats booked
- ii). The total number of economy class seats booked = the number of economy class seats which are occupied + the number of economy class seats which remained vacant
- iii). The total number of business class seats booked = the number of business class seats which are occupied + the number of business class seats which remained vacant

iv). In each flight, all the booked seats are confirmed in both economy and business class, unless mentioned in the question.

In IndiGo, the number of seats occupied by the females in business class is 40% of the number of seats occupied by the females in economy class of the same flight. If the number of occupied by the males in economy and business class are equal, then find the ratio of the number of seats occupied by males and females in economy class in IndiGo. (The seats are occupied by males and females only)

- a. 7:5
- b. 3:2
- c. 2:1
- d. Cannot be determined
- e. None of these

12. Questions

Quantity I: If the number of seats which are not booked in economy class in Vistara is $[X+20]$, which is $[1/(X-95)]$ of the number of seats booked in the economy class of the same flight. Find the value of $[X/5]$. (X is a positive integer)

Quantity II: If the total number of seats available in business class in Vistara is $[8Y]$, out of which 37.50% seats are not booked. Find the value of $[Y-40]$.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

13. Questions

The total number of seats booked in another flight Air India is 30% less than Vistara and the total number of business class seats booked in Air India is $[Z]$, which is equal to the number of seats which are occupied in economy class in the same flight. In Air India, if the number of seats which remained vacant in business class is 20 less than economy class and the number of seats which are occupied in business class is 160, then find the value of $[\sqrt{(Z-55)}]$.

- a. 12
- b. 18
- c. 15
- d. 11
- e. None of these

14. Questions

In SpiceJet, if the ratio of the number of Indians who occupied their seats in economy class and foreigners who occupied their seats in business class is 19:4, while the number of foreigners who occupied their seats in economy class is equal to the number of Indians who occupied their seats in business class, then find the total number of Indians who occupied their seats (economy + business) in SpiceJet. (One person can occupy only one seat)

- a. 175
- b. 250
- c. 180
- d. 215
- e. None of these

15. Questions

In IndiGo, out of the total number of seats booked in economy class, if $[T]\%$ seats are further cancelled, so that the number of seats which are occupied becomes $[25T]$. If the number of seats which remained vacant is unchanged, then find the number of seats which are cancelled in economy class in IndiGo.

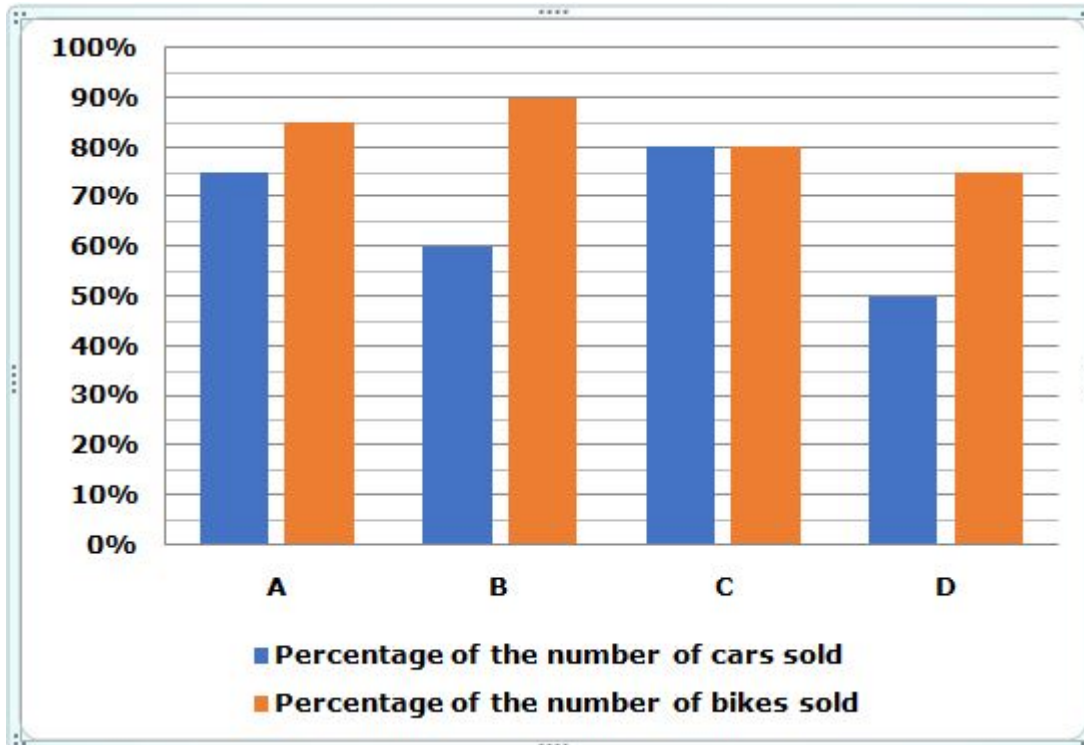
- a. 24
- b. 18
- c. 50
- d. 36
- e. None of these

16. Questions

Read the following information carefully and answer the questions based on it.

The bar graph given below shows the percentage of the number of cars sold out of the total number cars manufactured (sold + unsold) by four different showrooms A, B, C and D. It also shows the percentage of the number of bikes sold out of the total number of bikes manufactured (sold + unsold) by each showroom.

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Note:

- i). The total number of vehicles manufactured = the number of cars manufactured + the number of bikes manufactured
- ii). The total number of cars manufactured = the number of cars sold + the number of cars unsold
- iii). The total number of bikes manufactured = the number of bikes sold + the number of bikes unsold
- iv). The total number of vehicles manufactured by showroom A, B, C and D is 360, 300, 250 and 200 respectively.
- v). The ratio of the number of cars manufactured by showroom A and the number of bikes manufactured by showroom B is 12:5.
- vi). The number of bikes manufactured by showroom A is $\frac{3}{5}$ th of the number of cars manufactured by showroom B.
- vii). The difference between the number of cars and bikes unsold by showroom C is 20. (cars > bikes)
- viii). Out of the total number of vehicles manufactured by showroom D, 42.50% remained unsold.

The ratio of the number of cars sold to males by showroom B and the number of bikes sold to females by the same showroom B is 5:2, while the number of cars sold to females by showroom B is 15 less than the number of bikes sold to males by showroom B.

X = The difference between the number of cars sold to males and females by showroom B.

Y = The percentage of the number of bikes sold to females by showroom B out of the total number of bikes sold by the same showroom.

- a. $X > Y$
- b. $X < Y$

- c. $X \geq Y$
- d. $X \leq Y$
- e. $X = Y$

17. Questions

Out of the total number of bikes manufactured by showroom C, the number of defective bikes is [Z], while rest of the bikes are non-defective. If the number of non-defective bikes which are sold is 60, while the number of defective bikes unsold is 5 less than the number of non-defective unsold, then find the number of defective bikes manufactured by showroom C. (Only non-defective bikes can be sold)

- a. 5
- b. 12
- c. 15
- d. 8
- e. None of these

18. Questions

In showroom A, if the cost price of each car is Rs. [50T], which is sold at a profit a 50% after allowing a discount Rs. 30000 on each car, then the overall selling price of all cars sold by showroom A will be Rs. 27000000. Find the mark-up percentage of each car manufactured by showroom A.

- a. 60%
- b. 75%
- c. 80%
- d. Cannot be determined
- e. None of these

19. Questions

The number of cars unsold by another showroom E is 80, which is 6.67% more than the number of bikes sold by the same showroom, while the number of bikes unsold by showroom E is $\frac{1}{4}$ th of the total number of bikes manufactured by the same showroom.

Which of the following can be determined?

- I). The total number of vehicles (cars + bikes) manufactured by showroom E.
- II). The ratio of the number of cars and bikes sold by showroom E.
- III). The difference between the number of bikes sold and unsold by showroom E.

- a. Both II and III
- b. Only III
- c. Only I
- d. All I, II and III
- e. None of these

20. Questions

Out of the total number of cars manufactured by showroom D, the number of petrol cars manufactured is ____[?]**_____** more than the number of electric cars manufactured. If the ratio of the number of diesel and electric cars manufactured is 2:1, then the number of petrol cars manufactured by showroom D becomes 40 more than the diesel cars manufactured by the same showroom. (Only petrol, diesel and electric cars are manufactured by showroom D)

What should come in place of question mark?

- a. 60
- b. 42
- c. 35
- d. 26
- e. None of these

Explanations:

1. Questions

Company P,

The total number of employees working = 450

The total number of employees working in HR department = $\frac{4}{9}$ of 450 = 200 (Note: $44.44\% = \frac{4}{9}$)

The total number of employees working in marketing department = $(450 - 200) = 250$

Let the number of male employees working in marketing and HR department be 15a and 14a respectively.

So, the number of female employees working in marketing department = $(250 - 15a)$

And, the number of female employees working in HR department = $(200 - 14a)$

Now,

$$(250 - 15a) / (200 - 14a) = \frac{5}{3}$$

$$750 - 45a = 1000 - 70a$$

$$25a = 250$$

$$a = 10$$

Therefore,

$$\text{The number of male employees working in marketing department} = 15a = (15 * 10) = 150$$

$$\text{The number of male employees working in HR department} = 14a = (14 * 10) = 140$$

$$\text{The number of female employees working in marketing department} = (250 - 15a) = (250 - 150) = 100$$

$$\text{The number of female employees working in HR department} = (200 - 14a) = (200 - 140) = 60$$

Company Q,

$$\text{The total number of employees working} = 600$$

$$\text{The total number of employees working in marketing department} = \{(600 + 160)/2\} = (760/2) = 380$$

$$\text{The total number of employees working in HR department} = \{(600 - 160)/2\} = (440/2) = 220$$

Let the number of male employees working in marketing and HR department be $26b$ and $17b$ respectively.

$$\text{So, the number of female employees working in marketing department} = (380 - 26b)$$

$$\text{And, the number of female employees working in HR department} = (220 - 17b)$$

Now,

$$(380 - 26b)/(220 - 17b) = 12/5$$

$$1900 - 130b = 2640 - 204b$$

$$74b = 740$$

$$b = 10$$

Therefore,

$$\text{The number of male employees working in marketing department} = 26b = (26 * 10) = 260$$

$$\text{The number of male employees working in HR department} = 17b = (17 * 10) = 170$$

$$\text{The number of female employees working in marketing department} = (380 - 26b) = (380 - 260) = 120$$

$$\text{The number of female employees working in HR department} = (220 - 17b) = (220 - 170) = 50$$

Company R and Company S,

$$\text{The total number of employees working in company R} = 280$$

$$\text{And, the total number of employees working in company S} = 360$$

The ratio of total number of employees working in marketing department in company R and S = 100:120 = 5:6. Let it be $5c$ and $6c$ respectively.

$$\text{So, the total number of employees working in HR department in company R} = (280 - 5c)$$

$$\text{And, the total number of employees working in HR department in company S} = (360 - 6c)$$

Now,

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$$(280-5c)/(360-6c) = 2/3$$

$$840 - 15c = 720 - 12c$$

$$3c = 120$$

$$c = 40$$

Company R

The total number of employees working = 280

The total number of employees working in marketing department = $5c = (5 * 40) = 200$

The total number of employees working in HR department = $(280-200) = 80$

Now, let the number of male employees working in marketing and HR department be $32d$ and $9d$ respectively.

So, the number of female employees working in marketing department = $(200-32d)$

And, the number of female employees working in HR department = $(80-9d)$

Now,

$$(200-32d)/(80-9d) = 8/7$$

$$1400 - 224d = 640 - 72d$$

$$152d = 760$$

$$d = 5$$

Therefore,

The number of male employees working in marketing department = $32d = (32 * 5) = 160$

The number of male employees working in HR department = $9d = (9 * 5) = 45$

The number of female employees working in marketing department = $(200-32d) = (200-160) = 40$

The number of female employees working in HR department = $(80-9d) = (80-45) = 35$

Company S,

The total number of employees working = 360

The total number of employees working in marketing department = $6c = (6 * 40) = 240$

The total number of employees working in HR department = $(360-240) = 120$

Now, let the number of male employees working in marketing and HR department be $15e$ and $8e$ respectively.

So, the number of female employees working in marketing department = $(240-15e)$

And, the number of female employees working in HR department = $(120-8e)$

Now,

$$(240-15e)/(120-8e) = 9/4$$

$$960 - 60e = 1080 - 72e$$

$$12e = 120$$

$$e = 10$$

Therefore,

The number of male employees working in marketing department = $15e = (15 * 10) = 150$

The number of male employees working in HR department = $8e = (8 * 10) = 80$

The number of female employees working in marketing department = $(240-15e) = (240-150) = 90$

The number of female employees working in HR department = $(120-8e) = (120-80) = 40$

Marketing

Company	The total number of employees working (males + females)	The total number of male employees working	The total number of female employees working
P	250	150	100
Q	380	260	120
R	200	160	40
S	240	150	90
Total	1070	720	350

HR

Company	The total number of employees working (males + females)	The total number of male employees working	The total number of female employees working
P	200	140	60
Q	220	170	50
R	80	45	35
S	120	80	40
Total	620	435	185

Answer: B

Now, according to the question,

Company S (HR department)

The total number of male employees working = 80

The total number of female employees working = 40

Using I

The number of under graduate male employees working = 75% of 80 = 60

So, the number of under graduate female employees working = $(60-30) = 30$

Now, the number of post-graduate male employees working = $(80-60) = 20$

And, the number of post-graduate female employees working = $(40-30) = 10$

Therefore, the ratio of the number of post-graduate male and female employees working = $20:10 = 2:1$.

So, I does satisfy.

Using II

The number of under graduate male employees working = 55% of 80 = 44

So, the number of under graduate female employees working = $(44-22) = 22$

Now, the number of post-graduate male employees working = $(80-44) = 36$

And, the number of post-graduate female employees working = $(40-22) = 18$

Therefore, the ratio of the number of post-graduate male and female employees working = $36:18 = 2:1$.

So, II does satisfy.

Using III

The number of under graduate male employees working = 60% of 80 = 48

So, the number of under graduate female employees working = $(48-28) = 20$

Now, the number of post-graduate male employees working = $(80-48) = 32$

And, the number of post-graduate female employees working = $(40-20) = 20$

Therefore, the ratio of the number of post-graduate male and female employees working = $32:20 = 8:5$.

So, III does not satisfy.

Both I and II follow.

Hence, the correct answer is option B.

2. Questions

Company P,

The total number of employees working = 450

The total number of employees working in HR department = $\frac{4}{9}$ of 450 = 200 (**Note: $44.44\% = \frac{4}{9}$**)

The total number of employees working in marketing department = $(450-200) = 250$

Let the number of male employees working in marketing and HR department be $15a$ and $14a$ respectively.

So, the number of female employees working in marketing department = $(250-15a)$

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And, the number of female employees working in HR department = $(200-14a)$

Now,

$$(250-15a)/(200-14a) = 5/3$$

$$750 - 45a = 1000 - 70a$$

$$25a = 250$$

$$a = 10$$

Therefore,

$$\text{The number of male employees working in marketing department} = 15a = (15 * 10) = 150$$

$$\text{The number of male employees working in HR department} = 14a = (14 * 10) = 140$$

$$\text{The number of female employees working in marketing department} = (250-15a) = (250-150) = 100$$

$$\text{The number of female employees working in HR department} = (200-14a) = (200-140) = 60$$

Company Q,

$$\text{The total number of employees working} = 600$$

$$\text{The total number of employees working in marketing department} = \{(600+160)/2\} = (760/2) = 380$$

$$\text{The total number of employees working in HR department} = \{(600-160)/2\} = (440/2) = 220$$

Let the number of male employees working in marketing and HR department be $26b$ and $17b$ respectively.

$$\text{So, the number of female employees working in marketing department} = (380-26b)$$

$$\text{And, the number of female employees working in HR department} = (220-17b)$$

Now,

$$(380-26b)/(220-17b) = 12/5$$

$$1900 - 130b = 2640 - 204b$$

$$74b = 740$$

$$b = 10$$

Therefore,

$$\text{The number of male employees working in marketing department} = 26b = (26 * 10) = 260$$

$$\text{The number of male employees working in HR department} = 17b = (17 * 10) = 170$$

$$\text{The number of female employees working in marketing department} = (380-26b) = (380-260) = 120$$

$$\text{The number of female employees working in HR department} = (220-17b) = (220-170) = 50$$

Company R and Company S,

$$\text{The total number of employees working in company R} = 280$$

$$\text{And, the total number of employees working in company S} = 360$$

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The ratio of total number of employees working in marketing department in company R and S = 100:120 = 5:6. Let it be $5c$ and $6c$ respectively.

So, the total number of employees working in HR department in company R = $(280-5c)$

And, the total number of employees working in HR department in company S = $(360-6c)$

Now,

$$(280-5c)/(360-6c) = 2/3$$

$$840 - 15c = 720 - 12c$$

$$3c = 120$$

$$c = 40$$

Company R

The total number of employees working = 280

The total number of employees working in marketing department = $5c = (5 * 40) = 200$

The total number of employees working in HR department = $(280-200) = 80$

Now, let the number of male employees working in marketing and HR department be $32d$ and $9d$ respectively.

So, the number of female employees working in marketing department = $(200-32d)$

And, the number of female employees working in HR department = $(80-9d)$

Now,

$$(200-32d)/(80-9d) = 8/7$$

$$1400 - 224d = 640 - 72d$$

$$152d = 760$$

$$d = 5$$

Therefore,

The number of male employees working in marketing department = $32d = (32 * 5) = 160$

The number of male employees working in HR department = $9d = (9 * 5) = 45$

The number of female employees working in marketing department = $(200-32d) = (200-160) = 40$

The number of female employees working in HR department = $(80-9d) = (80-45) = 35$

Company S,

The total number of employees working = 360

The total number of employees working in marketing department = $6c = (6 * 40) = 240$

The total number of employees working in HR department = $(360-240) = 120$

Now, let the number of male employees working in marketing and HR department be $15e$ and $8e$

respectively.

So, the number of female employees working in marketing department = $(240-15e)$

And, the number of female employees working in HR department = $(120-8e)$

Now,

$$(240-15e)/(120-8e) = 9/4$$

$$960 - 60e = 1080 - 72e$$

$$12e = 120$$

$$e = 10$$

Therefore,

The number of male employees working in marketing department = $15e = (15 * 10) = 150$

The number of male employees working in HR department = $8e = (8 * 10) = 80$

The number of female employees working in marketing department = $(240-15e) = (240-150) = 90$

The number of female employees working in HR department = $(120-8e) = (120-80) = 40$

Marketing

Company	The total number of employees working (males + females)	The total number of male employees working	The total number of female employees working
P	250	150	100
Q	380	260	120
R	200	160	40
S	240	150	90
Total	1070	720	350

HR

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Company	The total number of employees working (males + females)	The total number of male employees working	The total number of female employees working
P	200	140	60
Q	220	170	50
R	80	45	35
S	120	80	40
Total	620	435	185

Answer: C

Company R (marketing department)

The total number of male employees working = 160

The total number of female employees working = 40

Let the number of male employees working who did not get promotions and female employees working who get promotions be 12m and 5m respectively.

The number of male employees working who get promotions = (160-12m)

And, the number of female employees working who did not get promotions = (40-5m)

Now,

$$(160-12m) - (40-5m) = 85$$

$$75 - 12m = 40 - 5m$$

$$7m = 35$$

$$m = 5$$

So, the total number of employees (males + females) who get promotions in marketing department of company R = $\{(160-12m)+5m\} = (160-7m) = (160-(7*5)) = (160-35) = 125$.

Hence, the correct answer is option C.

3. Questions

Company P,

The total number of employees working = 450

The total number of employees working in HR department = $\frac{4}{9}$ of 450 = 200 (**Note: 44.44% = $\frac{4}{9}$**)

The total number of employees working in marketing department = $(450-200) = 250$

Let the number of male employees working in marketing and HR department be 15a and 14a respectively.

So, the number of female employees working in marketing department = $(250-15a)$

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And, the number of female employees working in HR department = $(200-14a)$

Now,

$$(250-15a)/(200-14a) = 5/3$$

$$750 - 45a = 1000 - 70a$$

$$25a = 250$$

$$a = 10$$

Therefore,

$$\text{The number of male employees working in marketing department} = 15a = (15 * 10) = 150$$

$$\text{The number of male employees working in HR department} = 14a = (14 * 10) = 140$$

$$\text{The number of female employees working in marketing department} = (250-15a) = (250-150) = 100$$

$$\text{The number of female employees working in HR department} = (200-14a) = (200-140) = 60$$

Company Q,

$$\text{The total number of employees working} = 600$$

$$\text{The total number of employees working in marketing department} = \{(600+160)/2\} = (760/2) = 380$$

$$\text{The total number of employees working in HR department} = \{(600-160)/2\} = (440/2) = 220$$

Let the number of male employees working in marketing and HR department be $26b$ and $17b$ respectively.

$$\text{So, the number of female employees working in marketing department} = (380-26b)$$

$$\text{And, the number of female employees working in HR department} = (220-17b)$$

Now,

$$(380-26b)/(220-17b) = 12/5$$

$$1900 - 130b = 2640 - 204b$$

$$74b = 740$$

$$b = 10$$

Therefore,

$$\text{The number of male employees working in marketing department} = 26b = (26 * 10) = 260$$

$$\text{The number of male employees working in HR department} = 17b = (17 * 10) = 170$$

$$\text{The number of female employees working in marketing department} = (380-26b) = (380-260) = 120$$

$$\text{The number of female employees working in HR department} = (220-17b) = (220-170) = 50$$

Company R and Company S,

$$\text{The total number of employees working in company R} = 280$$

$$\text{And, the total number of employees working in company S} = 360$$

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The ratio of total number of employees working in marketing department in company R and S = 100:120 = 5:6. Let it be $5c$ and $6c$ respectively.

So, the total number of employees working in HR department in company R = $(280-5c)$

And, the total number of employees working in HR department in company S = $(360-6c)$

Now,

$$(280-5c)/(360-6c) = 2/3$$

$$840 - 15c = 720 - 12c$$

$$3c = 120$$

$$c = 40$$

Company R

The total number of employees working = 280

The total number of employees working in marketing department = $5c = (5 * 40) = 200$

The total number of employees working in HR department = $(280-200) = 80$

Now, let the number of male employees working in marketing and HR department be $32d$ and $9d$ respectively.

So, the number of female employees working in marketing department = $(200-32d)$

And, the number of female employees working in HR department = $(80-9d)$

Now,

$$(200-32d)/(80-9d) = 8/7$$

$$1400 - 224d = 640 - 72d$$

$$152d = 760$$

$$d = 5$$

Therefore,

The number of male employees working in marketing department = $32d = (32 * 5) = 160$

The number of male employees working in HR department = $9d = (9 * 5) = 45$

The number of female employees working in marketing department = $(200-32d) = (200-160) = 40$

The number of female employees working in HR department = $(80-9d) = (80-45) = 35$

Company S,

The total number of employees working = 360

The total number of employees working in marketing department = $6c = (6 * 40) = 240$

The total number of employees working in HR department = $(360-240) = 120$

Now, let the number of male employees working in marketing and HR department be $15e$ and $8e$

respectively.

So, the number of female employees working in marketing department = $(240-15e)$

And, the number of female employees working in HR department = $(120-8e)$

Now,

$$(240-15e)/(120-8e) = 9/4$$

$$960 - 60e = 1080 - 72e$$

$$12e = 120$$

$$e = 10$$

Therefore,

The number of male employees working in marketing department = $15e = (15 * 10) = 150$

The number of male employees working in HR department = $8e = (8 * 10) = 80$

The number of female employees working in marketing department = $(240-15e) = (240-150) = 90$

The number of female employees working in HR department = $(120-8e) = (120-80) = 40$

Marketing

Company	The total number of employees working (males + females)	The total number of male employees working	The total number of female employees working
P	250	150	100
Q	380	260	120
R	200	160	40
S	240	150	90
Total	1070	720	350

HR

@QuizByMickey

Company	The total number of employees working (males + females)	The total number of male employees working	The total number of female employees working
P	200	140	60
Q	220	170	50
R	80	45	35
S	120	80	40
Total	620	435	185

Answer: A

Company P

The total number of employees working in HR department = 200

The total number of employees working in sales department = 150% of 200 = 300

The number of female employees working in accounts department = 60

So, the number of female employees working in sales department = $(60 \times 2) = 120$

The number of male employees working in sales department = $(300 - 120) = 180$

So, the number of male employees working in accounts department = 180

Therefore, the total number of employees (males + females) working in accounts department = $(180 + 60) = 240$

Hence, the correct answer is option A.

4. Questions

Company P,

The total number of employees working = 450

The total number of employees working in HR department = $\frac{4}{9}$ of 450 = 200 (**Note: 44.44% = $\frac{4}{9}$**)

The total number of employees working in marketing department = $(450 - 200) = 250$

Let the number of male employees working in marketing and HR department be $15a$ and $14a$ respectively.

So, the number of female employees working in marketing department = $(250 - 15a)$

And, the number of female employees working in HR department = $(200 - 14a)$

Now,

$$(250 - 15a) / (200 - 14a) = \frac{5}{3}$$

$$750 - 45a = 1000 - 70a$$

$$25a = 250$$

$$a = 10$$

Therefore,

$$\text{The number of male employees working in marketing department} = 15a = (15 * 10) = 150$$

$$\text{The number of male employees working in HR department} = 14a = (14 * 10) = 140$$

$$\text{The number of female employees working in marketing department} = (250 - 15a) = (250 - 150) = 100$$

$$\text{The number of female employees working in HR department} = (200 - 14a) = (200 - 140) = 60$$

Company Q,

$$\text{The total number of employees working} = 600$$

$$\text{The total number of employees working in marketing department} = \{(600 + 160)/2\} = (760/2) = 380$$

$$\text{The total number of employees working in HR department} = \{(600 - 160)/2\} = (440/2) = 220$$

Let the number of male employees working in marketing and HR department be $26b$ and $17b$ respectively.

$$\text{So, the number of female employees working in marketing department} = (380 - 26b)$$

$$\text{And, the number of female employees working in HR department} = (220 - 17b)$$

Now,

$$(380 - 26b)/(220 - 17b) = 12/5$$

$$1900 - 130b = 2640 - 204b$$

$$74b = 740$$

$$b = 10$$

Therefore,

$$\text{The number of male employees working in marketing department} = 26b = (26 * 10) = 260$$

$$\text{The number of male employees working in HR department} = 17b = (17 * 10) = 170$$

$$\text{The number of female employees working in marketing department} = (380 - 26b) = (380 - 260) = 120$$

$$\text{The number of female employees working in HR department} = (220 - 17b) = (220 - 170) = 50$$

Company R and Company S,

$$\text{The total number of employees working in company R} = 280$$

$$\text{And, the total number of employees working in company S} = 360$$

The ratio of total number of employees working in marketing department in company R and S = 100:120 = 5:6. Let it be $5c$ and $6c$ respectively.

$$\text{So, the total number of employees working in HR department in company R} = (280 - 5c)$$

$$\text{And, the total number of employees working in HR department in company S} = (360 - 6c)$$

Now,

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$$(280-5c)/(360-6c) = 2/3$$

$$840 - 15c = 720 - 12c$$

$$3c = 120$$

$$c = 40$$

Company R

The total number of employees working = 280

The total number of employees working in marketing department = $5c = (5 * 40) = 200$

The total number of employees working in HR department = $(280-200) = 80$

Now, let the number of male employees working in marketing and HR department be $32d$ and $9d$ respectively.

So, the number of female employees working in marketing department = $(200-32d)$

And, the number of female employees working in HR department = $(80-9d)$

Now,

$$(200-32d)/(80-9d) = 8/7$$

$$1400 - 224d = 640 - 72d$$

$$152d = 760$$

$$d = 5$$

Therefore,

The number of male employees working in marketing department = $32d = (32 * 5) = 160$

The number of male employees working in HR department = $9d = (9 * 5) = 45$

The number of female employees working in marketing department = $(200-32d) = (200-160) = 40$

The number of female employees working in HR department = $(80-9d) = (80-45) = 35$

Company S,

The total number of employees working = 360

The total number of employees working in marketing department = $6c = (6 * 40) = 240$

The total number of employees working in HR department = $(360-240) = 120$

Now, let the number of male employees working in marketing and HR department be $15e$ and $8e$ respectively.

So, the number of female employees working in marketing department = $(240-15e)$

And, the number of female employees working in HR department = $(120-8e)$

Now,

$$(240-15e)/(120-8e) = 9/4$$

$$960 - 60e = 1080 - 72e$$

$$12e = 120$$

$$e = 10$$

Therefore,

The number of male employees working in marketing department = $15e = (15 * 10) = 150$

The number of male employees working in HR department = $8e = (8 * 10) = 80$

The number of female employees working in marketing department = $(240 - 15e) = (240 - 150) = 90$

The number of female employees working in HR department = $(120 - 8e) = (120 - 80) = 40$

Marketing

Company	The total number of employees working (males + females)	The total number of male employees working	The total number of female employees working
P	250	150	100
Q	380	260	120
R	200	160	40
S	240	150	90
Total	1070	720	350

HR

Company	The total number of employees working (males + females)	The total number of male employees working	The total number of female employees working
P	200	140	60
Q	220	170	50
R	80	45	35
S	120	80	40
Total	620	435	185

Answer: D

The total number of working in marketing department of company S = 240

Company T

Marketing Department

The total number of employees working = $5/3$ of $240 = 400$ (Note: $66.67\% = 2/3$)

The number of female employees working = $3/8$ of $400 = 150$

So, the number of male employees working = $5/8$ of $400 = 250$

HR Department

The number of male employees working = $(250/2) = 125$

And, the number of female employees working = $\{(150/2)*1\} = 75$

The total number of employees working = $(125+75) = 200$

Therefore, the total number of employees (marketing + HR) working in company T = $(400+200) = 600$.

Hence, the correct answer is option D.

5. Questions

Company P,

The total number of employees working = 450

The total number of employees working in HR department = $4/9$ of $450 = 200$ (Note: $44.44\% = 4/9$)

The total number of employees working in marketing department = $(450-200) = 250$

Let the number of male employees working in marketing and HR department be $15a$ and $14a$ respectively.

So, the number of female employees working in marketing department = $(250-15a)$

And, the number of female employees working in HR department = $(200-14a)$

Now,

$$(250-15a)/(200-14a) = 5/3$$

$$750 - 45a = 1000 - 70a$$

$$25a = 250$$

$$a = 10$$

Therefore,

The number of male employees working in marketing department = $15a = (15 * 10) = 150$

The number of male employees working in HR department = $14a = (14 * 10) = 140$

The number of female employees working in marketing department = $(250-15a) = (250-150) = 100$

The number of female employees working in HR department = $(200-14a) = (200-140) = 60$

Company Q,

The total number of employees working = 600

The total number of employees working in marketing department = $\{(600+160)/2\} = (760/2) = 380$

The total number of employees working in HR department = $\{(600-160)/2\} = (440/2) = 220$

Let the number of male employees working in marketing and HR department be $26b$ and $17b$ respectively.

So, the number of female employees working in marketing department = $(380-26b)$

And, the number of female employees working in HR department = $(220-17b)$

Now,

$$(380-26b)/(220-17b) = 12/5$$

$$1900 - 130b = 2640 - 204b$$

$$74b = 740$$

$$b = 10$$

Therefore,

The number of male employees working in marketing department = $26b = (26 * 10) = 260$

The number of male employees working in HR department = $17b = (17 * 10) = 170$

The number of female employees working in marketing department = $(380-26b) = (380-260) = 120$

The number of female employees working in HR department = $(220-17b) = (220-170) = 50$

Company R and Company S,

The total number of employees working in company R = 280

And, the total number of employees working in company S = 360

The ratio of total number of employees working in marketing department in company R and S = 100:120 = 5:6. Let it be $5c$ and $6c$ respectively.

So, the total number of employees working in HR department in company R = $(280-5c)$

And, the total number of employees working in HR department in company S = $(360-6c)$

Now,

$$(280-5c)/(360-6c) = 2/3$$

$$840 - 15c = 720 - 12c$$

$$3c = 120$$

$$c = 40$$

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Company R

The total number of employees working = 280

The total number of employees working in marketing department = $5c = (5 * 40) = 200$

The total number of employees working in HR department = $(280-200) = 80$

Now, let the number of male employees working in marketing and HR department be $32d$ and $9d$ respectively.

So, the number of female employees working in marketing department = $(200-32d)$

And, the number of female employees working in HR department = $(80-9d)$

Now,

$$(200-32d)/(80-9d) = 8/7$$

$$1400 - 224d = 640 - 72d$$

$$152d = 760$$

$$d = 5$$

Therefore,

$$\text{The number of male employees working in marketing department} = 32d = (32 * 5) = 160$$

$$\text{The number of male employees working in HR department} = 9d = (9 * 5) = 45$$

$$\text{The number of female employees working in marketing department} = (200-32d) = (200-160) = 40$$

$$\text{The number of female employees working in HR department} = (80-9d) = (80-45) = 35$$

Company S,

$$\text{The total number of employees working} = 360$$

$$\text{The total number of employees working in marketing department} = 6c = (6 * 40) = 240$$

$$\text{The total number of employees working in HR department} = (360-240) = 120$$

Now, let the number of male employees working in marketing and HR department be $15e$ and $8e$ respectively.

$$\text{So, the number of female employees working in marketing department} = (240-15e)$$

$$\text{And, the number of female employees working in HR department} = (120-8e)$$

Now,

$$(240-15e)/(120-8e) = 9/4$$

$$960 - 60e = 1080 - 72e$$

$$12e = 120$$

$$e = 10$$

Therefore,

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$$\text{The number of male employees working in marketing department} = 15e = (15 * 10) = 150$$

$$\text{The number of male employees working in HR department} = 8e = (8 * 10) = 80$$

$$\text{The number of female employees working in marketing department} = (240-15e) = (240-150) = 90$$

$$\text{The number of female employees working in HR department} = (120-8e) = (120-80) = 40$$

Marketing

Company	The total number of employees working (males + females)	The total number of male employees working	The total number of female employees working
P	250	150	100
Q	380	260	120
R	200	160	40
S	240	150	90
Total	1070	720	350

HR

Company	The total number of employees working (males + females)	The total number of male employees working	The total number of female employees working
P	200	140	60
Q	220	170	50
R	80	45	35
S	120	80	40
Total	620	435	185

Answer: B

Company Q (HR department)

The total number of male employees working initially = 170

The total number of female employees working initially = 50

Now, the number of newly joined male employees = $\frac{3}{13}$ of $65X = 15X$

And, the number of newly joined female employees = $\frac{10}{13}$ of $65X = 50X$

So,

$$(170+15X) - (50+50X) = 25X$$

$$170 - 10X = 50 + 50X$$

$$60X = 120$$

$$X = 2$$

2 is a prime number.

Therefore, ? = prime

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Hence, the correct answer is option B.

6. Questions

Answer: D

Using statements I, II and III together

Let the initial quantity of milk and water in vessel A be $2x$ ml and x ml respectively.

Also, let the initial quantity of milk and water in vessel B be $3y$ ml and $2y$ ml respectively.

Now, after mixing both the vessels A and B,

The total quantity of milk in the resultant mixture = $(2x+3y)$ ml

The total quantity of water in the resultant mixture = $(x+2y)$ ml

So,

$$(2x+3y) - (x+2y) = 116$$

$$x + y = 116 \dots\dots\dots(i)$$

Also, the total initial mixture of vessels A and B together = 420 ml

$$(2x+x) + (3y+2y) = 420$$

$$3x + 5y = 420 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$3x + 3y = 348$$

$$3x + 5y = 420$$

$$2y = 72$$

$$y = 36$$

Now, put the value of y in equation (i), we get,

$$x + y = 116$$

$$x + 36 = 116$$

$$x = 80$$

So, the initial mixture in vessel A = $(2x+x) = 3x = (3*80) = 240$ ml

And, the initial mixture in vessel B = $(3y+2y) = 5y = (5*36) = 180$ ml

Therefore, the difference between the initial mixture of vessel A and vessel B = $(240-180) = 60$ ml.

So, all the statements I, II and III together are necessary.

Hence, the correct answer is option D.

7. Questions

Answer: D

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Using statements I, II and III together

Article A

The C.P. = Rs. 8P

The S.P. = $11/8$ of 8P = Rs. 11P (Note: $37.50\% = 3/8$)

The M.P. = Rs. (11P+100)

Article B

The M.P. = Rs. (11P+100)

The S.P. = Rs. 960

The C.P. = {960-(3P+60)} = Rs. (900-3P)

Now,

$$(900-3P)/(11P+100) = 100/200$$

$$(900-3P)/(11P+100) = 1/2$$

$$1800 - 6P = 11P + 100$$

$$17P = 1700$$

$$P = 100$$

So, the marked price of article B = (11P+100) = (11*100+100) = (1100+100) = Rs. 1200.

$$900 - 3P = 900 - 3 * 100 = 600$$

Therefore, all the statements I, II and III together are necessary.

Hence, the correct answer is option D.

8. Questions

Answer: A

Using statements I and III together

Let the length of train X be L m

So, the length of train Y = (L+40) m

The length of the platform = 80 meters

The speed of train Y = 80 m/sec

So, the speed of train X = {(80/8)×9} = 90 m/sec (Note: $11.11\% = 1/9$)

Now,

$$\{(L+40+80)/80\} - \{(L+80)/90\} = 1$$

$$\{(L+120)/80\} - \{(L+80)/90\} = 1$$

$$\{(9L+1080)-(8L+640)\}/720 = 1$$

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$$L + 440 = 720$$

$$L = 280$$

So, the length of train Y = $(L+40) = (280+40) = 320$ meters

Therefore, both statements I and III together are necessary.

Using statements II and III together

Let the length of train X be L m.

Let the length of train Y be K m.

The speed of train Y = 80 m/sec

So, the speed of train X = $\{(80/8) \times 9\} = 90$ m/sec (**Note: 11.11% = 1/9**)

Now,

$$(90-20) = L/4$$

$$L = 280$$

Then,

$$\{(K+80)/80\} - \{(280+80)/90\} = 1$$

$$\{(K+80)/80\} - 4 = 1$$

$$(K+80)/80 = 5$$

$$K+80 = 400$$

$$K = 320$$

Hence, the correct answer is option A.

9. Questions

Answer: C

The amount invested in,

Scheme A (S.I.) = Rs. 32000

Scheme B (C.I.) = $\{(32000/2) \times 3\} = \text{Rs. } 48000$

Using statements I and II together

Let the rate of interest offered in scheme B be R% per annum compounded annually.

The interest received from scheme B = Rs. 10080

So,

$$\{(10080/48000) \times 100\} = 21\%$$

$$[(R+R) + \{(R \times R)/100\}] = \{2R + (R^2/100)\} = \{(200R + R^2)/100\}$$

Now,

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$$\{(200R+R^2)/100\} = 21$$

$$R^2 + 200R = 2100$$

$$R^2 + 200R - 2100 = 0$$

$$R^2 + 210R - 10R - 2100 = 0$$

$$R(R+210) - 10(R+210) = 0$$

$$(R-10)(R+210) = 0$$

$$R = +10 \text{ or } -210$$

So, $R = 10$ (positive integer)

The rate of interest offered in scheme B = $R\% = 10\%$

So, the rate of the offered in scheme A = $\{(10/4) \times 5\}\% = 12.50\%$ p.a.

Now, the interest received from scheme A = $(12.50 \times 3)\%$ of 32000 = 37.50% of 32000 = Rs. 12000.

Using statements I and III together

Let the rate of interest offered in scheme B be $T\%$ per annum compounded annually.

The interest received from scheme B = Rs. 10080

So,

$$\{(10080/48000) \times 100\} = 21\%$$

$$[(T+T) + \{(T \times T)/100\}] = \{2T + (T^2/100)\} = \{(200T + T^2)/100\}$$

Now,

$$\{(200R+T^2)/100\} = 21$$

$$T^2 + 200T = 2100$$

$$T^2 + 200T - 2100 = 0$$

$$T^2 + 210T - 10T - 2100 = 0$$

$$T(T+210) - 10(T+210) = 0$$

$$(T-10)(T+210) = 0$$

$$T = +10 \text{ or } -210$$

So, $T = 10$ (positive integer)

The rate of interest offered in scheme B = $T\% = 10\%$

Now, the rate of interest offered in scheme A = $(10+2.50)\% = 12.50\%$ p.a.

Therefore, the interest received from scheme A = $(12.50 \times 3)\%$ of 32000 = 37.50% of 32000 = Rs. 12000.

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Using statements II and III together

The rate of interest offered in scheme B = $S\% = (2.5 \times 4)\% = 10\%$ p.a.

So, the rate of the offered in scheme A = $\{(10/4) \times 5\}\% = 12.50\%$ p.a.

Now, the interest received from scheme A = $(12.50 \times 3)\%$ of 32000 = 37.50% of 32000 = Rs. 12000.

So, Any of the two given statements are sufficient

Hence, the correct answer is option C.

10. Questions

Answer: D

Using statements I, II and III together

The total work = 60 units

The time taken to complete the whole work by,

$$(P+Q+R) = (60/13) \text{ days}$$

$$(Q+R) = (3 \times 2) = 6 \text{ days}$$

So, the efficiency of

$$(P+Q+R) = \{60/(60/13)\} = 13$$

$$(Q+R) = \{60/6\} = 10$$

$$\text{So, } P = (13-10) = 3$$

$$\text{And, } R = \{(3/3) \times 4\} = 4$$

Therefore, the time taken by R to complete the whole work alone = $(60/4) = 15$ days.

So, all statements I, II and III together are necessary.

Hence, the correct answer is option D.

11. Questions

IndiGo

The total number of seats booked = 800

The total number of economy class seats booked = $5/8$ of 800 = 500

The total number of business class seats booked = $3/8$ of 800 = 300

The ratio of the number of economy class seats which are occupied and business class seats which remained vacant = 4:1. Let it be $4a$ and a respectively.

Also, the ratio of the number business class seats which are occupied and economy class seats which remained vacant = 9:8. Let it be $9b$ and $8b$ respectively. (**Note: $12.50\% = 1/8$**)

Now,

$$4a + 8b = 500$$

$$a + 2b = 125 \dots\dots\dots(I)$$

And,

$$a + 9b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$7b = 175$$

$$b = 25$$

Now, put the value of b in equation (I), we get,

$$a + 2b = 125$$

$$a + (2 \times 25) = 125$$

$$a = 125 - 50$$

$$a = 75$$

Therefore,

$$\text{The number of economy class seats which are occupied} = 4a = (4 \times 75) = 300$$

$$\text{The number of economy class seats which remained vacant} = 8b = (8 \times 25) = 200$$

$$\text{The number of business class seats which are occupied} = 9b = (9 \times 25) = 225$$

$$\text{The number of business class seats which remained vacant} = a = 75$$

SpiceJet

$$\text{The total number of business class seats booked} = 150$$

$$\text{The total number of economy class seats booked} = \{(150/1) \times 3\} = 450$$

$$\text{So, the total number of seats booked} = (450 + 150) = 600$$

Let the number of economy class seats which are occupied and business class seats which remained vacant be $5c$ and c respectively.

Also, the ratio the number of business class seats which are occupied and economy class seats which remained vacant = 1:2. Let it be d and $2d$ respectively.

Now,

$$5c + 2d = 450 \dots\dots\dots(III)$$

$$c + d = 150 \dots\dots\dots(IV)$$

By solving equations (III) and (IV), we get,

$$5c + 2d = 450$$

$$5c + 5d = 750$$

$$3d = 300$$

$$d = 100$$

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Now, put the value of d in equation (III), we get,

$$5c + 2d = 450$$

$$5c + (2 \times 100) = 450$$

$$5c = 450 - 200$$

$$5c = 250$$

$$c = 50$$

Therefore,

$$\text{The number of economy class seats which are occupied} = 5c = (5 \times 50) = 250$$

$$\text{The number of economy class seats which remained vacant} = 2d = (2 \times 100) = 200$$

$$\text{The number of business class seats which are occupied} = d = 100$$

$$\text{The number of business class seats which remained vacant} = c = 50$$

Vistara

Let the number of economy class seats which are occupied and vacant be $2e$ and e respectively.

Also, the ratio of the number of business class seats which are occupied and vacant = 7:1. Let it be $7f$ and f respectively. (**Note: 14.28% = 1/7**)

Now,

$$2e - 7f = 50 \dots\dots\dots(V)$$

$$e - f = 150 \dots\dots\dots(VI)$$

By solving equations (V) and (VI), we get,

$$2e - 7f = 50$$

$$2e - 2f = 300$$

$$5f = 250$$

$$f = 50$$

Now, put the value of f in equation (V), we get,

$$2e - 7f = 50$$

$$2e - (7 \times 50) = 50$$

$$2e = 50 + 350$$

$$2e = 400$$

$$e = 200$$

Therefore,

$$\text{The number of economy class seats which are occupied} = 2e = (2 \times 200) = 400$$

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The number of economy class seats which remained vacant = $e = 200$

The total number of economy class seats booked = $(400+200) = 600$

The number of business class seats which are occupied = $7f = (7 \times 50) = 350$

The number of business class seats which remained vacant = $f = 50$

The total number of business class seats booked = $(350+50) = 400$

The total number of seats booked = $(600+400) = 1000$

Economy Class

Name of the flight	The total number of seats booked	The number of seats which are occupied	The number of seats which remained vacant
IndiGo	500	300	200
SpiceJet	450	250	200
Vistara	600	400	200
Total	1550	950	600

Business Class

Name of the flight	The total number of seats booked	The number of seats which are occupied	The number of seats which remained vacant
IndiGo	300	225	75
SpiceJet	150	100	50
Vistara	400	350	50
Total	850	675	175

Answer: A

Now, according to the question,

IndiGo,

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The total number of seats occupied (males + females) in economy class = 300

The total number of seats occupied (males + females) in business class = 225

Let the number of seats occupied by the males in economy and business class be m each.

So, the number of seats occupied by the females in economy class = $(300-m)$

And, the number of seats occupied by the females in business class = $(225-m)$

Now,

$$(225-m)/(300-m) = 40/100$$

$$(225-m)/(300-m) = 2/5$$

$$1125 - 5m = 600 - 2m$$

$$3m = 525$$

$$m = 175$$

Therefore, the number of seats occupied by the males in economy class = $m = 175$

And, the number of seats occupied by the females in economy class = $(300-m) = (300-175) = 125$

So, the required ratio = $175:125 = 7:5$

Hence, the correct answer is option A.

12. Questions

IndiGo

The total number of seats booked = 800

The total number of economy class seats booked = $5/8$ of $800 = 500$

The total number of business class seats booked = $3/8$ of $800 = 300$

The ratio of the number of economy class seats which are occupied and business class seats which remained vacant = $4:1$. Let it be $4a$ and a respectively.

Also, the ratio of the number business class seats which are occupied and economy class seats which remained vacant = $9:8$. Let it be $9b$ and $8b$ respectively. (Note: $12.50\% = 1/8$)

Now,

$$4a + 8b = 500$$

$$a + 2b = 125 \dots\dots\dots(I)$$

And,

$$a + 9b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$7b = 175$$

$$b = 25$$

Now, put the value of b in equation (I), we get,

$$a + 2b = 125$$

$$a + (2 \times 25) = 125$$

$$a = 125 - 50$$

$$a = 75$$

Therefore,

@QuizByMickey

The number of economy class seats which are occupied = $4a = (4 \times 75) = 300$

The number of economy class seats which remained vacant = $8b = (8 \times 25) = 200$

The number of business class seats which are occupied = $9b = (9 \times 25) = 225$

The number of business class seats which remained vacant = $a = 75$

SpiceJet

The total number of business class seats booked = 150

The total number of economy class seats booked = $\{(150/1) \times 3\} = 450$

So, the total number of seats booked = $(450 + 150) = 600$

Let the number of economy class seats which are occupied and business class seats which remained vacant be $5c$ and c respectively.

Also, the ratio the number of business class seats which are occupied and economy class seats which remained vacant = 1:2. Let it be d and $2d$ respectively.

Now,

$$5c + 2d = 450 \dots\dots\dots(III)$$

$$c + d = 150 \dots\dots\dots(IV)$$

By solving equations (III) and (IV), we get,

$$5c + 2d = 450$$

$$5c + 5d = 750$$

$$3d = 300$$

$$d = 100$$

Now, put the value of d in equation (III), we get,

$$5c + 2d = 450$$

$$5c + (2 \times 100) = 450$$

$$5c = 450 - 200$$

$$5c = 250$$

$$c = 50$$

Therefore,

The number of economy class seats which are occupied = $5c = (5 \times 50) = 250$

The number of economy class seats which remained vacant = $2d = (2 \times 100) = 200$

The number of business class seats which are occupied = $d = 100$

The number of business class seats which remained vacant = $c = 50$

Vistara

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Let the number of economy class seats which are occupied and vacant be $2e$ and e respectively.

Also, the ratio of the number of business class seats which are occupied and vacant = $7:1$. Let it be $7f$ and f respectively. (**Note: $14.28\% = 1/7$**)

Now,

$$2e - 7f = 50 \dots\dots\dots(V)$$

$$e - f = 150 \dots\dots\dots(VI)$$

By solving equations (V) and (VI), we get,

$$2e - 7f = 50$$

$$2e - 2f = 300$$

$$5f = 250$$

$$f = 50$$

Now, put the value of f in equation (V), we get,

$$2e - 7f = 50$$

$$2e - (7 \times 50) = 50$$

$$2e = 50 + 350$$

$$2e = 400$$

$$e = 200$$

Therefore,

$$\text{The number of economy class seats which are occupied} = 2e = (2 \times 200) = 400$$

$$\text{The number of economy class seats which remained vacant} = e = 200$$

$$\text{The total number of economy class seats booked} = (400 + 200) = 600$$

$$\text{The number of business class seats which are occupied} = 7f = (7 \times 50) = 350$$

$$\text{The number of business class seats which remained vacant} = f = 50$$

$$\text{The total number of business class seats booked} = (350 + 50) = 400$$

$$\text{The total number of seats booked} = (600 + 400) = 1000$$

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Economy Class

Name of the flight	The total number of seats booked	The number of seats which are occupied	The number of seats which remained vacant
IndiGo	500	300	200
SpiceJet	450	250	200
Vistara	600	400	200
Total	1550	950	600

Business Class

Name of the flight	The total number of seats booked	The number of seats which are occupied	The number of seats which remained vacant
IndiGo	300	225	75
SpiceJet	150	100	50
Vistara	400	350	50
Total	850	675	175

Answer: B

Vistara

The total number of seats booked in economy class = 600

The total number of seats booked in business class = 400

Quantity I

The total number of seats not booked in economy class = $(X+20)$

So,

$$(X+20)/600 = \{1/(X-95)\}$$

$$X^2 - 95X + 20X - 1900 = 600$$

$$X^2 - 75X - 2500 = 0$$

$$X^2 - 100X + 25X - 2500 = 0$$

$$X(X-100) + 25(X-100) = 0$$

$$(X+25)(X-100) = 0$$

$$X = -25 \text{ or } +100$$

So, $X = 100$ (positive integer)

$$\text{Quantity I} = (X/5) = (100/5) = 20.$$

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Quantity II

The total number of seats available = $8Y$

The number of seats which are not booked = 37.50% of $8Y = 3Y$

The total number of seats booked = $8Y - 3Y = 5Y$

Now,

$$5Y = 400$$

$$Y = 80$$

$$\text{Quantity II} = (Y - 40) = (80 - 40) = 40.$$

Therefore, Quantity I < Quantity II.

Hence, the correct answer is option B.

13. Questions

IndiGo

The total number of seats booked = 800

The total number of economy class seats booked = $\frac{5}{8}$ of 800 = 500

The total number of business class seats booked = $\frac{3}{8}$ of 800 = 300

The ratio of the number of economy class seats which are occupied and business class seats which remained vacant = 4:1. Let it be $4a$ and a respectively.

Also, the ratio of the number business class seats which are occupied and economy class seats which remained vacant = 9:8. Let it be $9b$ and $8b$ respectively. (Note: $12.50\% = \frac{1}{8}$)

Now,

$$4a + 8b = 500$$

$$a + 2b = 125 \dots\dots\dots(I)$$

And,

$$a + 9b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$7b = 175$$

$$b = 25$$

Now, put the value of b in equation (I), we get,

$$a + 2b = 125$$

$$a + (2 \times 25) = 125$$

$$a = 125 - 50$$

$$a = 75$$

@QuizByMickey

Therefore,

The number of economy class seats which are occupied = $4a = (4 \times 75) = 300$

The number of economy class seats which remained vacant = $8b = (8 \times 25) = 200$

The number of business class seats which are occupied = $9b = (9 \times 25) = 225$

The number of business class seats which remained vacant = $a = 75$

SpiceJet

The total number of business class seats booked = 150

The total number of economy class seats booked = $\{(150/1) \times 3\} = 450$

So, the total number of seats booked = $(450 + 150) = 600$

Let the number of economy class seats which are occupied and business class seats which remained vacant be $5c$ and c respectively.

Also, the ratio the number of business class seats which are occupied and economy class seats which remained vacant = 1:2. Let it be d and $2d$ respectively.

Now,

$$5c + 2d = 450 \dots\dots\dots(III)$$

$$c + d = 150 \dots\dots\dots(IV)$$

By solving equations (III) and (IV), we get,

$$5c + 2d = 450$$

$$5c + 5d = 750$$

$$3d = 300$$

$$d = 100$$

Now, put the value of d in equation (III), we get,

$$5c + 2d = 450$$

$$5c + (2 \times 100) = 450$$

$$5c = 450 - 200$$

$$5c = 250$$

$$c = 50$$

Therefore,

The number of economy class seats which are occupied = $5c = (5 \times 50) = 250$

The number of economy class seats which remained vacant = $2d = (2 \times 100) = 200$

The number of business class seats which are occupied = $d = 100$

The number of business class seats which remained vacant = $c = 50$

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Vistara

Let the number of economy class seats which are occupied and vacant be $2e$ and e respectively.

Also, the ratio of the number of business class seats which are occupied and vacant = $7:1$. Let it be $7f$ and f respectively. (**Note: $14.28\% = 1/7$**)

Now,

$$2e - 7f = 50 \dots\dots\dots(V)$$

$$e - f = 150 \dots\dots\dots(VI)$$

By solving equations (V) and (VI), we get,

$$2e - 7f = 50$$

$$2e - 2f = 300$$

$$5f = 250$$

$$f = 50$$

Now, put the value of f in equation (V), we get,

$$2e - 7f = 50$$

$$2e - (7 \times 50) = 50$$

$$2e = 50 + 350$$

$$2e = 400$$

$$e = 200$$

Therefore,

The number of economy class seats which are occupied = $2e = (2 \times 200) = 400$

The number of economy class seats which remained vacant = $e = 200$

The total number of economy class seats booked = $(400 + 200) = 600$

The number of business class seats which are occupied = $7f = (7 \times 50) = 350$

The number of business class seats which remained vacant = $f = 50$

The total number of business class seats booked = $(350 + 50) = 400$

The total number of seats booked = $(600 + 400) = 1000$

Economy Class

@QuizByMickey

Name of the flight	The total number of seats booked	The number of seats which are occupied	The number of seats which remained vacant
IndiGo	500	300	200
SpiceJet	450	250	200
Vistara	600	400	200
Total	1550	950	600

Business Class

Name of the flight	The total number of seats booked	The number of seats which are occupied	The number of seats which remained vacant
IndiGo	300	225	75
SpiceJet	150	100	50
Vistara	400	350	50
Total	850	675	175

Answer: C

The total number of seats booked in Vistara = 1000

Air India

The total number of seats booked = 70% of 1000 = 700

The total number of seats booked in business class = Z

So, the total number of seats booked in economy class = (700-Z)

The number of seats which are occupied in economy class = Z

So, the number of seats which remained vacant in economy class = $\{(700-Z)-Z\} = (700-2Z)$

And, the number of seats which remained vacant in business class = $\{(700-2Z)-20\} = (680-2Z)$

Now, the number of seats which are occupied in business class = 160

So,

$$160 + (680-2Z) = Z$$

$$3Z = 840$$

$$Z = 280$$

Therefore, the value of $\sqrt{(Z-55)} = \sqrt{(280-55)} = \sqrt{225} = 15$.

Hence, the correct answer is option C.

14. Questions

IndiGo

The total number of seats booked = 800

The total number of economy class seats booked = $\frac{5}{8}$ of 800 = 500

The total number of business class seats booked = $\frac{3}{8}$ of 800 = 300

The ratio of the number of economy class seats which are occupied and business class seats which remained vacant = 4:1. Let it be $4a$ and a respectively.

Also, the ratio of the number business class seats which are occupied and economy class seats which remained vacant = 9:8. Let it be $9b$ and $8b$ respectively. (Note: $12.50\% = \frac{1}{8}$)

Now,

$$4a + 8b = 500$$

$$a + 2b = 125 \dots\dots\dots(I)$$

And,

$$a + 9b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$7b = 175$$

$$b = 25$$

Now, put the value of b in equation (I), we get,

$$a + 2b = 125$$

$$a + (2 \times 25) = 125$$

$$a = 125 - 50$$

$$a = 75$$

Therefore,

The number of economy class seats which are occupied = $4a = (4 \times 75) = 300$

The number of economy class seats which remained vacant = $8b = (8 \times 25) = 200$

The number of business class seats which are occupied = $9b = (9 \times 25) = 225$

The number of business class seats which remained vacant = $a = 75$

SpiceJet

The total number of business class seats booked = 150

The total number of economy class seats booked = $\{(150/1) \times 3\} = 450$

So, the total number of seats booked = $(450 + 150) = 600$

Let the number of economy class seats which are occupied and business class seats which remained vacant

be $5c$ and c respectively.

Also, the ratio the number of business class seats which are occupied and economy class seats which remained vacant = $1:2$. Let it be d and $2d$ respectively.

Now,

$$5c + 2d = 450 \dots\dots\dots(III)$$

$$c + d = 150 \dots\dots\dots(IV)$$

By solving equations (III) and (IV), we get,

$$5c + 2d = 450$$

$$5c + 5d = 750$$

$$3d = 300$$

$$d = 100$$

Now, put the value of d in equation (III), we get,

$$5c + 2d = 450$$

$$5c + (2 \times 100) = 450$$

$$5c = 450 - 200$$

$$5c = 250$$

$$c = 50$$

Therefore,

The number of economy class seats which are occupied = $5c = (5 \times 50) = 250$

The number of economy class seats which remained vacant = $2d = (2 \times 100) = 200$

The number of business class seats which are occupied = $d = 100$

The number of business class seats which remained vacant = $c = 50$

Vistara

Let the number of economy class seats which are occupied and vacant be $2e$ and e respectively.

Also, the ratio of the number of business class seats which are occupied and vacant = $7:1$. Let it be $7f$ and f respectively. (**Note: $14.28\% = 1/7$**)

Now,

$$2e - 7f = 50 \dots\dots\dots(V)$$

$$e - f = 150 \dots\dots\dots(VI)$$

By solving equations (V) and (VI), we get,

$$2e - 7f = 50$$

$$2e - 2f = 300$$

$$5f = 250$$

$$f = 50$$

Now, put the value of f in equation (V), we get,

$$2e - 7f = 50$$

$$2e - (7 \times 50) = 50$$

$$2e = 50 + 350$$

$$2e = 400$$

$$e = 200$$

Therefore,

The number of economy class seats which are occupied $= 2e = (2 \times 200) = 400$

The number of economy class seats which remained vacant $= e = 200$

The total number of economy class seats booked $= (400 + 200) = 600$

The number of business class seats which are occupied $= 7f = (7 \times 50) = 350$

The number of business class seats which remained vacant $= f = 50$

The total number of business class seats booked $= (350 + 50) = 400$

The total number of seats booked $= (600 + 400) = 1000$

Economy Class

Name of the flight	The total number of seats booked	The number of seats which are occupied	The number of seats which remained vacant
IndiGo	500	300	200
SpiceJet	450	250	200
Vistara	600	400	200
Total	1550	950	600

Business Class

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Name of the flight	The total number of seats booked	The number of seats which are occupied	The number of seats which remained vacant
IndiGo	300	225	75
SpiceJet	150	100	50
Vistara	400	350	50
Total	850	675	175

Answer: B

SpiceJet

The total number of seats occupied in economy class = 250

The total number of seats occupied in business class = 100

Let the number of Indians who occupied their seats in economy class and foreigners who occupied their seats in business class be $19p$ and $4p$ respectively.

So, the number of Indians who occupied their seats in business class = $(100-4p)$

And, the number of foreigners who occupied their seats in economy class = $(250-19p)$

Now,

$$(100-4p) = (250-19p)$$

$$15p = 150$$

$$p = 10$$

Therefore, the total number of Indians who occupied their seats (economy + business) in SpiceJet = $\{19p + (100-4p)\} = (15p + 100) = (15 \times 10 + 100) = (150 + 100) = 250$.

Hence, the correct answer is option B.

15. Questions

IndiGo

The total number of seats booked = 800

The total number of economy class seats booked = $\frac{5}{8}$ of 800 = 500

The total number of business class seats booked = $\frac{3}{8}$ of 800 = 300

The ratio of the number of economy class seats which are occupied and business class seats which remained vacant = 4:1. Let it be $4a$ and a respectively.

Also, the ratio of the number business class seats which are occupied and economy class seats which remained vacant = 9:8. Let it be $9b$ and $8b$ respectively. (**Note: $12.50\% = \frac{1}{8}$**)

Now,

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$$4a + 8b = 500$$

$$a + 2b = 125 \dots\dots\dots(I)$$

And,

$$a + 9b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$7b = 175$$

$$b = 25$$

Now, put the value of b in equation (I), we get,

$$a + 2b = 125$$

$$a + (2 \times 25) = 125$$

$$a = 125 - 50$$

$$a = 75$$

Therefore,

$$\text{The number of economy class seats which are occupied} = 4a = (4 \times 75) = 300$$

$$\text{The number of economy class seats which remained vacant} = 8b = (8 \times 25) = 200$$

$$\text{The number of business class seats which are occupied} = 9b = (9 \times 25) = 225$$

$$\text{The number of business class seats which remained vacant} = a = 75$$

SpiceJet

$$\text{The total number of business class seats booked} = 150$$

$$\text{The total number of economy class seats booked} = \{(150/1) \times 3\} = 450$$

$$\text{So, the total number of seats booked} = (450 + 150) = 600$$

Let the number of economy class seats which are occupied and business class seats which remained vacant be $5c$ and c respectively.

Also, the ratio the number of business class seats which are occupied and economy class seats which remained vacant = 1:2. Let it be d and $2d$ respectively.

@QuizByMickey

Now,

$$5c + 2d = 450 \dots\dots\dots(III)$$

$$c + d = 150 \dots\dots\dots(IV)$$

By solving equations (III) and (IV), we get,

$$5c + 2d = 450$$

$$5c + 5d = 750$$

$$3d = 300$$

$$d = 100$$

Now, put the value of d in equation (III), we get,

$$5c + 2d = 450$$

$$5c + (2 \times 100) = 450$$

$$5c = 450 - 200$$

$$5c = 250$$

$$c = 50$$

Therefore,

$$\text{The number of economy class seats which are occupied} = 5c = (5 \times 50) = 250$$

$$\text{The number of economy class seats which remained vacant} = 2d = (2 \times 100) = 200$$

$$\text{The number of business class seats which are occupied} = d = 100$$

$$\text{The number of business class seats which remained vacant} = c = 50$$

Vistara

Let the number of economy class seats which are occupied and vacant be $2e$ and e respectively.

Also, the ratio of the number of business class seats which are occupied and vacant = 7:1. Let it be $7f$ and f respectively. (**Note: 14.28% = 1/7**)

Now,

$$2e - 7f = 50 \dots\dots\dots(V)$$

$$e - f = 150 \dots\dots\dots(VI)$$

By solving equations (V) and (VI), we get,

$$2e - 7f = 50$$

$$2e - 2f = 300$$

$$5f = 250$$

$$f = 50$$

Now, put the value of f in equation (V), we get,

$$2e - 7f = 50$$

$$2e - (7 \times 50) = 50$$

$$2e = 50 + 350$$

$$2e = 400$$

$$e = 200$$

Therefore,

@QuizByMickey

The number of economy class seats which are occupied = $2e = (2 \times 200) = 400$

The number of economy class seats which remained vacant = $e = 200$

The total number of economy class seats booked = $(400+200) = 600$

The number of business class seats which are occupied = $7f = (7 \times 50) = 350$

The number of business class seats which remained vacant = $f = 50$

The total number of business class seats booked = $(350+50) = 400$

The total number of seats booked = $(600+400) = 1000$

Economy Class

Name of the flight	The total number of seats booked	The number of seats which are occupied	The number of seats which remained vacant
IndiGo	500	300	200
SpiceJet	450	250	200
Vistara	600	400	200
Total	1550	950	600

Business Class

Name of the flight	The total number of seats booked	The number of seats which are occupied	The number of seats which remained vacant
IndiGo	300	225	75
SpiceJet	150	100	50
Vistara	400	350	50
Total	850	675	175

Answer: C

IndiGo (economy class)

The total number of seats booked = 500

The number of seats cancelled = $T\%$ of 500 = $5T$

So, the number of seats confirmed = $(500-5T)$

The number of seats which remained vacant = 200

So, the number of seats which are occupied = $25T$

Now,

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$$200 + 25T = 500 - 5T$$

$$30T = 300$$

$$T = 10$$

Therefore, the number of seats which are cancelled in economy class in IndiGo = $5T = (5 \times 10) = 50$.

Hence, the correct answer is option C.

16. Questions

The total number of vehicles manufactured by showroom A = 360

The total number of vehicles manufactured by showroom B = 300

Now, let the number of cars manufactured by showroom A and bikes manufactured by showroom B be 12m and 5n respectively.

Also, the ratio of the number of bikes manufactured by showroom A and cars manufactured by showroom B = 3:5. Let it be 3n and 5n respectively.

Now,

$$12m + 3n = 360$$

$$4m + n = 120 \dots \dots \dots (I)$$

And,

$$5m + 5n = 300$$

$$m + n = 60 \dots \dots \dots (II)$$

By solving equations (I) and (II), we get,

$$4m + n = 120$$

$$4m + 4n = 240$$

$$3n = 120$$

$$n = 40$$

Now, put the value of n in equation (I), we get,

$$4m + n = 120$$

$$4m + 40 = 120$$

$$4m = 80$$

$$m = 20$$

Showroom A

The total number of vehicles manufactured = 360

The total number of cars manufactured = $12m = (12 \times 20) = 240$

The number of cars sold = 75% of 240 = 180

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The number of cars unsold = 25% of 240 = 60

The total number of bikes manufactured = $3n = (3 \times 40) = 120$

The number of bikes sold = 85% of 120 = 102

The number of bikes unsold = 15% of 120 = 18

Showroom B

The total number of vehicles manufactured = 300

The total number of cars manufactured = $5n = (5 \times 40) = 200$

The number of cars sold = 60% of 200 = 120

The number of cars unsold = 40% of 200 = 80

The total number of bikes manufactured = $5m = (5 \times 20) = 100$

The number of bikes sold = 90% of 100 = 90

The number of bikes unsold = 10% of 100 = 10

Showroom C

The total number of vehicles manufactured = 250

Let the total number of cars manufactured be k.

So, the total number of bikes manufactured = $(250 - k)$

Now, the number of cars sold = 80% of k = $0.80k$

And, the number of cars unsold = 20% of k = $0.20k$

The number of bikes sold = 80% of $(250 - k) = (200 - 0.80k)$

The number of bikes unsold = 20% of $(250 - k) = (50 - 0.20k)$

Now,

$$0.20k - (50 - 0.20k) = 20$$

$$0.20k - 20 = 50 - 0.20k$$

$$0.40k = 70$$

$$k = 175$$

Therefore,

The total number of cars manufactured = k = 175

The total number of bikes manufactured = $(250 - k) = (250 - 175) = 75$

The number of cars sold = $0.80k = (0.80 \times 175) = 140$

The number of cars unsold = $0.20k = (0.20 \times 175) = 35$

The number of bikes sold = $(200 - 0.80k) = (200 - 140) = 60$

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The number of bikes unsold = $(50 - 0.20k) = (50 - 35) = 15$

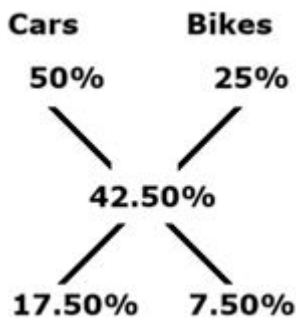
Showroom D

The percentage of the number of cars unsold = $(100 - 50)\% = 50\%$

The percentage of the number of bikes unsold = $(100 - 75)\% = 25\%$

The percentage of the total number of vehicles unsold = 42.50%

Using allegation method, we get,



So, the ratio of the total number of cars and bikes manufactured = $17.50\% : 7.50\% = 7:3$.

The total number of vehicles manufactured = 200

The total number cars manufactured = $7/10$ of $200 = 140$

The number of cars sold = 50% of $140 = 70$

The number of cars unsold = 50% of $140 = 70$

The total number bikes manufactured = $3/10$ of $200 = 60$

The number of bikes sold = 75% of $60 = 45$

The number of bikes unsold = 25% of $60 = 15$

Cars

Showrooms	The total number of cars manufactured	The number of cars sold	The number of cars unsold
A	240	180	60
B	200	120	80
C	175	140	35
D	140	70	70
Total	755	510	245

Bikes

Showrooms	The total number of bikes manufactured	The number of bikes sold	The number of bikes unsold
A	120	102	18
B	100	90	10
C	75	60	15
D	60	45	15
Total	355	297	58

Answer: B

Now, according to the question,

Showroom B

The total number of cars sold (males + females) = 120

The total number of bikes sold (males + females) = 90

Now, let the number of cars sold to males and bikes sold to females be 5p and 2p respectively.

So, the number of cars sold to females = (120-5p)

And, the number of bikes sold to males = (90-2p)

Now,

$$(90-2p) - (120-5p) = 15$$

$$75 - 2p = 120 - 5p$$

$$3p = 45$$

$$p = 15$$

$$X = \text{The number of cars sold to males} = 5p = (5 \times 15) = 75$$

$$\text{The number of cars sold to females} = (120-5p) = (120-75) = 45$$

$$\text{The required difference} = (75-45) = 30$$

$$\text{So, } X = 30$$

$$Y = \text{The number of bikes sold to females} = 2p = (2 \times 15) = 30$$

$$\text{The required percentage} = \{(30/90) \times 100\} = 33.33\%$$

$$\text{So, } Y = 33.33$$

Therefore, $X < Y$.

Hence, the correct answer is option B.

17. Questions

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The total number of vehicles manufactured by showroom A = 360

The total number of vehicles manufactured by showroom B = 300

Now, let the number of cars manufactured by showroom A and bikes manufactured by showroom B be 12m and 5m respectively.

Also, the ratio of the number of bikes manufactured by showroom A and cars manufactured by showroom B = 3:5. Let it be 3n and 5n respectively.

Now,

$$12m + 3n = 360$$

$$4m + n = 120 \dots\dots\dots(I)$$

And,

$$5m + 5n = 300$$

$$m + n = 60 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$4m + n = 120$$

$$4m + 4n = 240$$

$$3n = 120$$

$$n = 40$$

Now, put the value of n in equation (I), we get,

$$4m + n = 120$$

$$4m + 40 = 120$$

$$4m = 80$$

$$m = 20$$

Showroom A

The total number of vehicles manufactured = 360

The total number of cars manufactured = $12m = (12 \times 20) = 240$ @QuizByMickey

The number of cars sold = 75% of 240 = 180

The number of cars unsold = 25% of 240 = 60

The total number of bikes manufactured = $3n = (3 \times 40) = 120$

The number of bikes sold = 85% of 120 = 102

The number of bikes unsold = 15% of 120 = 18

Showroom B

The total number of vehicles manufactured = 300

The total number of cars manufactured = $5n = (5 \times 40) = 200$

The number of cars sold = 60% of 200 = 120

The number of cars unsold = 40% of 200 = 80

The total number of bikes manufactured = $5m = (5 \times 20) = 100$

The number of bikes sold = 90% of 100 = 90

The number of bikes unsold = 10% of 100 = 10

Showroom C

The total number of vehicles manufactured = 250

Let the total number of cars manufactured be k .

So, the total number of bikes manufactured = $(250 - k)$

Now, the number of cars sold = 80% of $k = 0.80k$

And, the number of cars unsold = 20% of $k = 0.20k$

The number of bikes sold = 80% of $(250 - k) = (200 - 0.80k)$

The number of bikes unsold = 20% of $(250 - k) = (50 - 0.20k)$

Now,

$$0.20k - (50 - 0.20k) = 20$$

$$0.20k - 20 = 50 - 0.20k$$

$$0.40k = 70$$

$$k = 175$$

Therefore,

The total number of cars manufactured = $k = 175$

The total number of bikes manufactured = $(250 - k) = (250 - 175) = 75$

The number of cars sold = $0.80k = (0.80 \times 175) = 140$

The number of cars unsold = $0.20k = (0.20 \times 175) = 35$

The number of bikes sold = $(200 - 0.80k) = (200 - 140) = 60$

The number of bikes unsold = $(50 - 0.20k) = (50 - 35) = 15$

Showroom D

The percentage of the number of cars unsold = $(100 - 50)\% = 50\%$

The percentage of the number of bikes unsold = $(100 - 75)\% = 25\%$

The percentage of the total number of vehicles unsold = 42.50%

Using allegation method, we get,

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So, the ratio of the total number of cars and bikes manufactured = $17.50\%:7.50\% = 7:3$.

The total number of vehicles manufactured = 200

The total number cars manufactured = $7/10$ of 200 = 140

The number of cars sold = 50% of 140 = 70

The number of cars unsold = 50% of 140 = 70

The total number bikes manufactured = $3/10$ of 200 = 60

The number of bikes sold = 75% of 60 = 45

The number of bikes unsold = 25% of 60 = 15

Cars

Showrooms	The total number of cars manufactured	The number of cars sold	The number of cars unsold
A	240	180	60
B	200	120	80
C	175	140	35
D	140	70	70
Total	755	510	245

Bikes

Showrooms	The total number of bikes manufactured	The number of bikes sold	The number of bikes unsold
A	120	102	18
B	100	90	10
C	75	60	15
D	60	45	15
Total	355	297	58

Answer: A

Showroom C

The total number of bikes manufactured = 75

The number of defective bikes manufactured = Z (all unsold)

So, the total number of non-defective bikes sold = (75-Z)

The number of non-defective bikes sold = 60

So, the number of non-defective bikes unsold = $\{(75-Z)-60\} = (15-Z)$

Now,

$$(15-Z) - Z = 5$$

$$15 - 2Z = 5$$

$$2Z = 10$$

$$Z = 5$$

Therefore, the number of defective manufactured by showroom C = $Z = 5$

Hence, the correct answer is option A.

18. Questions

The total number of vehicles manufactured by showroom A = 360

The total number of vehicles manufactured by showroom B = 300

Now, let the number of cars manufactured by showroom A and bikes manufactured by showroom B be 12m and 5m respectively.

Also, the ratio of the number of bikes manufactured by showroom A and cars manufactured by showroom B = 3:5. Let it be 3n and 5n respectively.

Now,

$$12m + 3n = 360$$

$$4m + n = 120 \dots \dots \dots (I)$$

And,

$$5m + 5n = 300$$

$$m + n = 60 \dots \dots \dots (II)$$

By solving equations (I) and (II), we get,

$$4m + n = 120$$

$$4m + 4n = 240$$

$$3n = 120$$

$$n = 40$$

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Now, put the value of n in equation (I), we get,

$$4m + n = 120$$

$$4m + 40 = 120$$

$$4m = 80$$

$$m = 20$$

Showroom A

The total number of vehicles manufactured = 360

The total number of cars manufactured = $12m = (12 \times 20) = 240$

The number of cars sold = 75% of 240 = 180

The number of cars unsold = 25% of 240 = 60

The total number of bikes manufactured = $3n = (3 \times 40) = 120$

The number of bikes sold = 85% of 120 = 102

The number of bikes unsold = 15% of 120 = 18

Showroom B

The total number of vehicles manufactured = 300

The total number of cars manufactured = $5n = (5 \times 40) = 200$

The number of cars sold = 60% of 200 = 120

The number of cars unsold = 40% of 200 = 80

The total number of bikes manufactured = $5m = (5 \times 20) = 100$

The number of bikes sold = 90% of 100 = 90

The number of bikes unsold = 10% of 100 = 10

Showroom C

The total number of vehicles manufactured = 250

Let the total number of cars manufactured be k .

So, the total number of bikes manufactured = $(250 - k)$

Now, the number of cars sold = 80% of $k = 0.80k$

And, the number of cars unsold = 20% of $k = 0.20k$

The number of bikes sold = 80% of $(250 - k) = (200 - 0.80k)$

The number of bikes unsold = 20% of $(250 - k) = (50 - 0.20k)$

Now,

$$0.20k - (50 - 0.20k) = 20$$

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$$0.20k - 20 = 50 - 0.20k$$

$$0.40k = 70$$

$$k = 175$$

Therefore,

The total number of cars manufactured = $k = 175$

The total number of bikes manufactured = $(250 - k) = (250 - 175) = 75$

The number of cars sold = $0.80k = (0.80 \times 175) = 140$

The number of cars unsold = $0.20k = (0.20 \times 175) = 35$

The number of bikes sold = $(200 - 0.80k) = (200 - 140) = 60$

The number of bikes unsold = $(50 - 0.20k) = (50 - 35) = 15$

Showroom D

The percentage of the number of cars unsold = $(100 - 50)\% = 50\%$

The percentage of the number of bikes unsold = $(100 - 75)\% = 25\%$

The percentage of the total number of vehicles unsold = 42.50%

Using allegation method, we get,



So, the ratio of the total number of cars and bikes manufactured = $17.50\% : 7.50\% = 7:3$.

The total number of vehicles manufactured = 200

The total number cars manufactured = $7/10$ of 200 = 140

The number of cars sold = 50% of 140 = 70

The number of cars unsold = 50% of 140 = 70

The total number bikes manufactured = $3/10$ of 200 = 60

The number of bikes sold = 75% of 60 = 45

The number of bikes unsold = 25% of 60 = 15

Cars

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Showrooms	The total number of cars manufactured	The number of cars sold	The number of cars unsold
A	240	180	60
B	200	120	80
C	175	140	35
D	140	70	70
Total	755	510	245

Bikes

Showrooms	The total number of bikes manufactured	The number of bikes sold	The number of bikes unsold
A	120	102	18
B	100	90	10
C	75	60	15
D	60	45	15
Total	355	297	58

Answer: C

Showroom A

The total number of cars manufactured = 240

The total number of cars sold = 180

The C.P. of each car = Rs. 50T

The S.P. of each car = 150% of 50T = Rs. 75T

The M.P. of each car = (75T+30000)

So, the overall S.P. all cars sold = (75T*180) = Rs. 13500T

Now,

$$13500T = 27000000$$

$$T = 2000$$

So, the M.P. of each car = (75T+30000) = (75*2000+30000) = (150000+30000) = Rs. 180000

And, the C.P. of each car = 50T = (50*2000) = Rs. 100000

Therefore, the mark-up percentage of each car = $\left[\frac{(180000-100000)}{100000}\right]\% = \left[\frac{(80000)}{100000} \times 100\right] = 80\%$.

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Hence, the correct answer is option C.

19. Questions

The total number of vehicles manufactured by showroom A = 360

The total number of vehicles manufactured by showroom B = 300

Now, let the number of cars manufactured by showroom A and bikes manufactured by showroom B be 12m and 5m respectively.

Also, the ratio of the number of bikes manufactured by showroom A and cars manufactured by showroom B = 3:5. Let it be 3n and 5n respectively.

Now,

$$12m + 3n = 360$$

$$4m + n = 120 \dots\dots\dots(I)$$

And,

$$5m + 5n = 300$$

$$m + n = 60 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$4m + n = 120$$

$$4m + 4n = 240$$

$$3n = 120$$

$$n = 40$$

Now, put the value of n in equation (I), we get,

$$4m + n = 120$$

$$4m + 40 = 120$$

$$4m = 80$$

$$m = 20$$

Showroom A

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The total number of vehicles manufactured = 360

The total number of cars manufactured = $12m = (12 \times 20) = 240$

The number of cars sold = 75% of 240 = 180

The number of cars unsold = 25% of 240 = 60

The total number of bikes manufactured = $3n = (3 \times 40) = 120$

The number of bikes sold = 85% of 120 = 102

The number of bikes unsold = 15% of 120 = 18

Showroom B

The total number of vehicles manufactured = 300

The total number of cars manufactured = $5n = (5 \times 40) = 200$

The number of cars sold = 60% of 200 = 120

The number of cars unsold = 40% of 200 = 80

The total number of bikes manufactured = $5m = (5 \times 20) = 100$

The number of bikes sold = 90% of 100 = 90

The number of bikes unsold = 10% of 100 = 10

Showroom C

The total number of vehicles manufactured = 250

Let the total number of cars manufactured be k .

So, the total number of bikes manufactured = $(250 - k)$

Now, the number of cars sold = 80% of $k = 0.80k$

And, the number of cars unsold = 20% of $k = 0.20k$

The number of bikes sold = 80% of $(250 - k) = (200 - 0.80k)$

The number of bikes unsold = 20% of $(250 - k) = (50 - 0.20k)$

Now,

$$0.20k - (50 - 0.20k) = 20$$

$$0.20k - 20 = 50 - 0.20k$$

$$0.40k = 70$$

$$k = 175$$

Therefore,

The total number of cars manufactured = $k = 175$

The total number of bikes manufactured = $(250 - k) = (250 - 175) = 75$

The number of cars sold = $0.80k = (0.80 \times 175) = 140$

The number of cars unsold = $0.20k = (0.20 \times 175) = 35$

The number of bikes sold = $(200 - 0.80k) = (200 - 140) = 60$

The number of bikes unsold = $(50 - 0.20k) = (50 - 35) = 15$

Showroom D

The percentage of the number of cars unsold = $(100 - 50)\% = 50\%$

The percentage of the number of bikes unsold = $(100 - 75)\% = 25\%$

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The percentage of the total number of vehicles unsold = 42.50%

Using allegation method, we get,



So, the ratio of the total number of cars and bikes manufactured = 17.50%:7.50% = 7:3.

The total number of vehicles manufactured = 200

The total number cars manufactured = 7/10 of 200 = 140

The number of cars sold = 50% of 140 = 70

The number of cars unsold = 50% of 140 = 70

The total number bikes manufactured = 3/10 of 200 = 60

The number of bikes sold = 75% of 60 = 45

The number of bikes unsold = 25% of 60 = 15

Cars

Showrooms	The total number of cars manufactured	The number of cars sold	The number of cars unsold
A	240	180	60
B	200	120	80
C	175	140	35
D	140	70	70
Total	755	510	245

Bikes

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Showrooms	The total number of bikes manufactured	The number of bikes sold	The number of bikes unsold
A	120	102	18
B	100	90	10
C	75	60	15
D	60	45	15
Total	355	297	58

Answer: B

Showroom E

The number of cars unsold = 80

The number of bikes sold = $\{(80/16) * 15\} = 75$ (Note: $6.67\% = 1/15$)

The ratio of the number of bikes sold and unsold = 3:1.

So, the number of bikes unsold = $\{(75/3) * 1\} = 25$

Therefore, the total number of bikes manufactured = $(75+25) = 100$

I). We don't have any adequate information from which we can determine the total number of cars manufactured.

So, I cannot be determined.

II). We don't have any adequate information from which we can determine the total number of cars sold.

So, II cannot be determined.

III). The number of bikes sold = 75

The number of bikes unsold = 25

The required difference = $(75-25) = 50$

So, III can be determined.

Therefore, only III can be determined.

Hence, the correct answer is option B.

20. Questions

The total number of vehicles manufactured by showroom A = 360

The total number of vehicles manufactured by showroom B = 300

Now, let the number of cars manufactured by showroom A and bikes manufactured by showroom B be 12m and 5m respectively.

Also, the ratio of the number of bikes manufactured by showroom A and cars manufactured by showroom

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B = 3:5. Let it be $3n$ and $5n$ respectively.

Now,

$$12m + 3n = 360$$

$$4m + n = 120 \dots\dots\dots(I)$$

And,

$$5m + 5n = 300$$

$$m + n = 60 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$4m + n = 120$$

$$4m + 4n = 240$$

$$3n = 120$$

$$n = 40$$

Now, put the value of n in equation (I), we get,

$$4m + n = 120$$

$$4m + 40 = 120$$

$$4m = 80$$

$$m = 20$$

Showroom A

The total number of vehicles manufactured = 360

The total number of cars manufactured = $12m = (12 \times 20) = 240$

The number of cars sold = 75% of 240 = 180

The number of cars unsold = 25% of 240 = 60

The total number of bikes manufactured = $3n = (3 \times 40) = 120$

The number of bikes sold = 85% of 120 = 102

The number of bikes unsold = 15% of 120 = 18

Showroom B

The total number of vehicles manufactured = 300

The total number of cars manufactured = $5n = (5 \times 40) = 200$

The number of cars sold = 60% of 200 = 120

The number of cars unsold = 40% of 200 = 80

The total number of bikes manufactured = $5m = (5 \times 20) = 100$

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The number of bikes sold = 90% of 100 = 90

The number of bikes unsold = 10% of 100 = 10

Showroom C

The total number of vehicles manufactured = 250

Let the total number of cars manufactured be k.

So, the total number of bikes manufactured = (250-k)

Now, the number of cars sold = 80% of k = 0.80k

And, the number of cars unsold = 20% of k = 0.20k

The number of bikes sold = 80% of (250-k) = (200-0.80k)

The number of bikes unsold = 20% of (250-k) = (50-0.20k)

Now,

$$0.20k - (50 - 0.20k) = 20$$

$$0.20k - 20 = 50 - 0.20k$$

$$0.40k = 70$$

$$k = 175$$

Therefore,

The total number of cars manufactured = k = 175

The total number of bikes manufactured = (250-k) = (250-175) = 75

The number of cars sold = 0.80k = (0.80×175) = 140

The number of cars unsold = 0.20k = (0.20×175) = 35

The number of bikes sold = (200-0.80k) = (200-140) = 60

The number of bikes unsold = (50-0.20k) = (50-35) = 15

Showroom D

The percentage of the number of cars unsold = (100-50)% = 50%

The percentage of the number of bikes unsold = (100-75)% = 25%

The percentage of the total number of vehicles unsold = 42.50%

Using allegation method, we get,

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So, the ratio of the total number of cars and bikes manufactured = $17.50\%:7.50\% = 7:3$.

The total number of vehicles manufactured = 200

The total number cars manufactured = $7/10$ of 200 = 140

The number of cars sold = 50% of 140 = 70

The number of cars unsold = 50% of 140 = 70

The total number bikes manufactured = $3/10$ of 200 = 60

The number of bikes sold = 75% of 60 = 45

The number of bikes unsold = 25% of 60 = 15

Cars

Showrooms	The total number of cars manufactured	The number of cars sold	The number of cars unsold
A	240	180	60
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D	140	70	70
Total	755	510	245

Bikes

Showrooms	The total number of bikes manufactured	The number of bikes sold	The number of bikes unsold
A	120	102	18
B	100	90	10
C	75	60	15
D	60	45	15
Total	355	297	58

Answer: A

Showroom D

The total number of cars manufactured = 140

Let the number of diesel and electric cars manufactured be $2v$ and v respectively.

So, the petrol cars manufactured = $\{140 - (2v + v)\} = (140 - 3v)$

Now,

$$(140 - 3v) - 2v = 40$$

$$140 - 5v = 40$$

$$5v = 100$$

$$v = 20$$

Now, the difference between the number of petrol and electric cars manufactured (petrol – electric) = $\{(140 - 3v) - v\} = (140 - 4v) = (140 - 4 \times 20) = (140 - 80) = 60$.

Therefore, ? = 60.

Hence, the correct answer is option A.

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